

# Topic 1: Independent Practice

## Covering Lessons 1-1 to SC-2

### Directions & Guidelines:

- **Show All Work in Notebook:** All work must be present in your spiral notebook. For word problems, you must show all work using the same **We Need, We Know, We Solve** framework taught in class.
- **If Receive Help:** Circle questions where you got stuck, referred to explanations, or where you received help. After help, solve each circled question correctly three times in a row.
- **How to Check Answers Automatically:** Explanations are available for the assignment, break each question down in extreme detail, and are accessible from the one-pager. Honestly attempt Part 1 before using explanations. You may check your answers on Canvas for accuracy using the link on the one-pager.
- **Completion & Memory Tracking:** Complete Part 1 before Part 2. When you complete the Independent Practice for a lesson (such as Lesson 1-1 or Lesson 1-2), add an entry for that lesson's Independent Practice to your memory sheet / spreadsheet. Do not have separate entries for Parts 1 and 2. Then, place the assignment in your binder.
- **Pacing:** Aim to complete a lesson's Independent Practice in one sitting. Students may work at their own pace and are free to work ahead.
- **Calculations & Answer Format:** No calculator. Write decimal answers as decimals and fraction answers in simplest terms. Round to two decimal places.

### Lesson 1-1: Relate Integers and Opposites

| Part 1                                                                                                                                                                                                                                        | Part 2                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1.</b> A submarine rises 18 m from a point below sea level, then descends 18 m.<br><b>• Part A:</b> Represent both changes with signed numbers and find their sum.<br><b>• Part B:</b> Explain why the two changes are additive inverses. | <b>Q1.</b> A hiker descends 24 m from a trail marker, then climbs 24 m.<br><b>• Part A:</b> Represent both changes with signed numbers and find their sum.<br><b>• Part B:</b> Explain why the two changes are additive inverses. |
| <b>Q2.</b> On a number line, zero is located in the middle. We have two mystery numbers, $a$ and $b$ , where $a < 0$ , $b > 0$ , and $a + b = 0$ . Write an equation about the absolute values of $a$ and $b$ that must be true.              | <b>Q2.</b> On a number line, zero is located in the middle. We have two mystery numbers, $p$ and $q$ , where $p > 0$ , $q < 0$ , and $p + q = 0$ . Write an equation about the absolute values of $p$ and $q$ that must be true.  |
| <b>Q3.</b> Evaluate: $ -9  =$                                                                                                                                                                                                                 | <b>Q3.</b> Evaluate: $ -11  =$                                                                                                                                                                                                    |
| <b>Q4.</b> Order from least to greatest: $-3, 8, -1$                                                                                                                                                                                          | <b>Q4.</b> Order from least to greatest: $6, -2, 4$                                                                                                                                                                               |

### Lesson 1-2: Understand Rational Numbers

| Part 1                                                                                                                                                                                                                                                                     | Part 2                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1.</b> In a survey, 3 out of 8 students chose basketball as their favorite sport. What is the decimal equivalent of the fraction of students who chose basketball?                                                                                                     | <b>Q1.</b> In a survey, 5 out of 8 students chose soccer as their favorite sport. What is the decimal equivalent of the fraction of students who chose soccer?                                                                                                                                                  |
| <b>Q2.</b> A student uses long division to convert a fraction $\frac{x}{y}$ (where $y \neq 0$ ) into a decimal. Which outcome is <b>NOT</b> possible?<br>A) The decimal terminates.<br>B) The decimal repeats a pattern.<br>C) The decimal neither terminates nor repeats. | <b>Q2.</b> A student uses long division to convert a fraction $\frac{a}{b}$ into a decimal. Which statement is <b>ALWAYS</b> right?<br>A) The decimal will eventually terminate.<br>B) The decimal will eventually terminate or repeat a pattern.<br>C) The quotient $\frac{a}{b}$ never terminates or repeats. |
| <b>Q3.</b> In a class debate, Jordan says 0.27 is rational because its digits repeat. Casey says it is not rational because it never ends. Which student is correct? Use a fraction or equation to support the claim.                                                      | <b>Q3.</b> In a class debate, Mia says $0.\overline{6}$ is rational because it equals $\frac{2}{3}$ . Lee says it is not rational because it never ends. Which student is correct? Use a fraction or equation to support the claim.                                                                             |

## Lesson SC-1: Compare Rational Numbers

| Part 1                                                                                                                                                                                       | Part 2                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1.</b> Your pet cat weighs 10.4 lbs. Your friend's rabbit weighs 10.6 lbs. Write two different statements using the less than (<) and greater than (>) symbols that compare the weights. | <b>Q1.</b> Your pet dog weighs 9.3 lbs. Your neighbor's cat weighs 9.8 lbs. Write two different statements using the less than (<) and greater than (>) symbols that compare the weights. |
| <b>Q2.</b> Compare the fraction $\frac{3}{4}$ with the decimal 0.75. Select all symbols that appropriately relate the two numbers (<, >, =, ≠).                                              | <b>Q2.</b> Compare the fraction $\frac{2}{5}$ with the decimal 0.4. Select all symbols that appropriately relate the two numbers (<, >, =, ≠).                                            |

## Lesson 1-3: Add Integers

| Part 1                                                                                                                                                                                                                  | Part 2                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1.</b> In round 1 of a game, a player scores 10 points, then loses 4 points, then loses 8 points.<br>• <b>Part A:</b> Explain how to show the change on a number line.<br>• <b>Part B:</b> What is the final score? | <b>Q1.</b> In round 1 of a game, a player scores 12 points, then loses 5 points, then loses 9 points.<br>• <b>Part A:</b> Explain how to show the change on a number line.<br>• <b>Part B:</b> What is the final score? |
| <b>Q2.</b> A student starts with 15 points. They lose 20 points in round 1 and gain 3 points in round 2. What is the final score?                                                                                       | <b>Q2.</b> A student starts with 18 points. They gain 4 points in round 1 and lose 25 points in round 2. What is the final score?                                                                                       |
| <b>Q3.</b> A bank account starts with \$500. The bank deducts a monthly maintenance fee of \$12.50. With no other activity, how much is in the account after 6 months?                                                  | <b>Q3.</b> A bank account starts with \$400. The bank deducts a monthly maintenance fee of \$8.75. With no other activity, how much is in the account after 8 months?                                                   |
| <b>Q4.</b> Evaluate: $-12 + 19 =$                                                                                                                                                                                       | <b>Q4.</b> Evaluate: $-8 + 14 =$                                                                                                                                                                                        |

## Lesson 1-4: Subtract Integers

| Part 1                                                                                                                                                                                                                             | Part 2                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1.</b> At 6:00 PM, the temperature was $5^{\circ}\text{F}$ . By midnight, it dropped to $-8^{\circ}\text{F}$ . Write at least two mathematical expressions that correctly represent the distance between the two temperatures. | <b>Q1.</b> At 7:00 AM, the temperature was $3^{\circ}\text{F}$ . By noon, it rose to $12^{\circ}\text{F}$ . Write at least two mathematical expressions that correctly represent the distance between the two temperatures. |
| <b>Q2.</b> Bird A is 150 ft above sea level. Bird B is 120 ft above sea level. Write at least two expressions for the vertical distance between them.                                                                              | <b>Q2.</b> Plane A is 280 ft above sea level. Plane B is 210 ft above sea level. Write at least two expressions for the vertical distance between them.                                                                     |
| <b>Q3.</b> Fish A is 25 ft below sea level ( $-25$ ft). Fish B is 40 ft below sea level ( $-40$ ft). Write at least two expressions for the vertical distance between them.                                                        | <b>Q3.</b> Submarine A is 18 ft below sea level ( $-18$ ft). Submarine B is 45 ft below sea level ( $-45$ ft). Write at least two expressions for the vertical distance between them.                                       |
| <b>Q4.</b> An elevator starts on floor -3, travels to floor 8, then returns to floor -1. What total distance does the elevator travel?                                                                                             | <b>Q4.</b> An elevator starts on floor -5, travels to floor 6, then returns to floor -2. What total distance does the elevator travel?                                                                                      |

## Lesson 1-5: Add and Subtract Rational Numbers

| Part 1                                                                                                                                                                        | Part 2                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1.</b> An account starts with \$200. The bank charges \$15.00 each month if the balance is below \$250. With no other activity, what is the balance after 4 months?       | <b>Q1.</b> An account starts with \$180. The bank charges \$12.00 each month if the balance is below \$200. With no other activity, what is the balance after 5 months?       |
| <b>Q2.</b> At 6:00 PM, the temperature is $-3.75^{\circ}\text{C}$ . It drops $2.4^{\circ}\text{C}$ , then drops $1.85^{\circ}\text{C}$ . What is the temperature at midnight? | <b>Q2.</b> At 4:00 PM, the temperature is $-4.6^{\circ}\text{C}$ . It drops $1.35^{\circ}\text{C}$ , then drops $2.05^{\circ}\text{C}$ . What is the temperature at midnight? |
| <b>Q3.</b> A lookout is $2\frac{1}{2}$ miles above sea level. A mountain base is 1.75 miles below sea level. What is the vertical distance between them?                      | <b>Q3.</b> A radio tower is $3\frac{1}{4}$ miles above sea level. A canyon floor is 2.5 miles below sea level. What is the vertical distance between them?                    |

## Lesson 1-6: Multiply Integers

| Part 1                                                                                                                                                                                                              | Part 2                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1.</b> A hiker begins at sea level (0 ft) and descends into a canyon at 15 ft/min.<br>• <b>Part A:</b> Unit rate of descent as an integer?<br>• <b>Part B:</b> Position relative to sea level after 12 minutes? | <b>Q1.</b> A climber begins at sea level (0 ft) and ascends a cliff at 20 ft/min.<br>• <b>Part A:</b> Unit rate of ascent as an integer?<br>• <b>Part B:</b> Position relative to sea level after 9 minutes? |
| <b>Q2.</b> An RC plane descends at 250.50 m/s. What is the change in height after $4\frac{1}{2}$ seconds? (Round to nearest kilometer).                                                                             | <b>Q2.</b> A hot-air balloon ascends at 180.25 m/s. What is the change in height after $3\frac{1}{2}$ seconds? (Round to nearest kilometer).                                                                 |
| <b>Q3.</b> Evaluate: $(-5)^2 =$                                                                                                                                                                                     | <b>Q3.</b> Evaluate: $(-6)^2 =$                                                                                                                                                                              |
| <b>Q4.</b> Evaluate: $-5^2 =$<br><i>Explain briefly how Q3 and Q4 differ.</i>                                                                                                                                       | <b>Q4.</b> Evaluate: $-6^2 =$<br><i>Explain briefly how Q3 and Q4 differ.</i>                                                                                                                                |

## Lesson 1-7: Multiply Rational Numbers

| Part 1                                | Part 2                                |
|---------------------------------------|---------------------------------------|
| <b>Q1.</b> Evaluate: $-0.5(10 + 6) =$ | <b>Q1.</b> Evaluate: $-0.2(15 + 5) =$ |

## Lesson 1-8: Divide Integers

| Part 1                                                                                                                                                                                                                         | Part 2                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1.</b> A submarine starts at sea level (0 ft) and descends to 168 ft in 7 minutes at a steady rate.<br>• <b>Part A:</b> Unit rate of descent (ft/min)?<br>• <b>Part B:</b> Position relative to sea level after 4 minutes? | <b>Q1.</b> A drone starts at sea level (0 ft) and climbs to 140 ft in 5 minutes at a steady rate.<br>• <b>Part A:</b> Unit rate of ascent (ft/min)?<br>• <b>Part B:</b> Position relative to sea level after 3 minutes? |
| <b>Q2.</b> Evaluate: $(-96) \div 8 =$                                                                                                                                                                                          | <b>Q2.</b> Evaluate: $(-81) \div 9 =$                                                                                                                                                                                   |
| <b>Q3.</b> Evaluate: $(-72) \div (-9) =$                                                                                                                                                                                       | <b>Q3.</b> Evaluate: $(-56) \div (-7) =$                                                                                                                                                                                |
| <b>Q4.</b> Find the value of $n$ : $\frac{-18}{6} = \frac{n}{-4}$                                                                                                                                                              | <b>Q4.</b> Find the value of $n$ : $\frac{-24}{8} = \frac{n}{-3}$                                                                                                                                                       |

## Lesson 1-9: Divide Rational Numbers

| Part 1                                                                                                                                                                  | Part 2                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1.</b> A business loses \$450.75 due to shipping delay. The 3 owners share the loss equally. Write the quotient representing each person's share.                   | <b>Q1.</b> A business loses \$625.50 due to damaged inventory. The 5 owners share the loss equally. Write the quotient representing each person's share.         |
| <b>Q2.</b> A race is $15\frac{1}{2}$ miles long. Water stations are placed every $3\frac{7}{8}$ miles and at the finish line. How many total water stations are needed? | <b>Q2.</b> A trail is $16\frac{1}{2}$ miles long. Rest stops are placed every $2\frac{3}{4}$ miles and at the finish line. How many total rest stops are needed? |

## Lesson 1-10: Solve Problems with Rational Numbers

| Part 1                                                                                                                                                               | Part 2                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1.</b> Hiker 1 climbs to a peak 450 ft above sea level. Hiker 2 descends into a valley 120 ft below sea level. What is the total vertical distance between them? | <b>Q1.</b> Hiker 1 climbs to a ridge 320 ft above sea level. Hiker 2 descends into a canyon 85 ft below sea level. What is the total vertical distance between them? |

## Lesson SC-2: Use the Laws of Exponents

| Part 1                                                                                                                                                                | Part 2                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1.</b> Poster area is given by $3^2 \times 3^4 \text{ in}^2$ . Simplify using exponent laws and find the total area.                                              | <b>Q1.</b> Banner area is given by $2^3 \times 2^5 \text{ in}^2$ . Simplify using exponent laws and find the total area.                                            |
| <b>Q2.</b> Made $2^4$ cupcakes, divided equally among $2^2$ friends.<br>• <b>Part A:</b> Expression for total per friend?<br>• <b>Part B:</b> How many cupcakes each? | <b>Q2.</b> Made $3^4$ muffins, divided equally among $3^2$ friends.<br>• <b>Part A:</b> Expression for total per friend?<br>• <b>Part B:</b> How many muffins each? |
| <b>Q3.</b> Evaluate: $(4^3)^2 =$                                                                                                                                      | <b>Q3.</b> Evaluate: $(2^4)^2 =$                                                                                                                                    |
| <b>Q4.</b> Evaluate: $7^0 =$                                                                                                                                          | <b>Q4.</b> Evaluate: $5^0 =$                                                                                                                                        |
| <b>Q5.</b> Simplify: $\frac{4^5}{4^2} \cdot 4^0 \cdot 4^{-1}$                                                                                                         | <b>Q5.</b> Simplify: $\frac{5^5}{5^2} \cdot 5^0 \cdot 5^{-1}$                                                                                                       |
| <b>Q6.</b> Simplify: $(2^3)^2 \cdot (3 \cdot 2)^2 \div \left(\frac{12}{2}\right)^2$                                                                                   | <b>Q6.</b> Simplify: $(3^2)^2 \cdot (2 \cdot 3)^2 \div \left(\frac{18}{3}\right)^2$                                                                                 |
| <b>Q7.</b> Simplify: $5^0 + 3^{-2} + \frac{2^6 \cdot 2^3}{2^7}$                                                                                                       | <b>Q7.</b> Simplify: $2^0 + 2^{-3}$                                                                                                                                 |
| <b>Q8.</b> Simplify: $\frac{7^4}{7^2} \cdot 7^0$                                                                                                                      | <b>Q8.</b> Simplify: $\frac{10^5}{10^3}$                                                                                                                            |
| <b>Q9.</b> Simplify: $(3^2)^3$                                                                                                                                        | <b>Q9.</b> Simplify: $(4^2)^2$                                                                                                                                      |
| <b>Q10.</b> Simplify: $(2 \cdot 5)^3$                                                                                                                                 | <b>Q10.</b> Simplify: $(2 \cdot 3)^2$                                                                                                                               |
| <b>Q11.</b> Simplify: $6^{-2} + \left(\frac{8}{4}\right)^3$                                                                                                           | <b>Q11.</b> Simplify: $\left(\frac{15}{3}\right)^2$                                                                                                                 |
| <b>Q12.</b> Simplify: $3^4 \cdot 3^2$                                                                                                                                 | <b>Q12.</b> Simplify: $5^3 \cdot 5^2$                                                                                                                               |